Statistics in Health Sciences - Assignment 1

G05: Judit Félelz Guerrero (17048330), Lídia Abio Rojo (1705251), Catalina Mascaró Català (1708159), Elia Campos Villaro (1703842)

28th October 2026

1 Introduction

The aim of this assignment is to apply the framework proposed by Basagaña et al. to the data set copd in the file copd redux.RData. Data copd is a subset of 45 variables of the original data analyzed in the cited paper.

2 Analysis

2.1 Description of missingness

Provide the same missingness description than in the original analysis. Specifically, reproduce the following text that is shown in the original paper:

> "The subset used here included 342 participants and 85 variables. The range of missing data in the different variables went from 0% to 47.7%, with 68.2% of the variables having less than 5% of the data missing. Only 13.7% of the participants had complete data on all 85 variables. Overall, 5.9% of the information was missing; that is, 5.9% of the total of 29,070 cells ($85 \times 342 = 29,070$) had missing data."

Instructions:

- Note that, since you are working with a subset of the original data set, red numbers above are not expected to be the same than in your results.
- In your `Rnw` document, the text above must be able to automatically update all red numbers in the case of changing the values of `copd`. I.e., if data change, then all red numbers must also automatically update accordingly in your `pdf` when compiling your `Rnw`. As a hint, you can follow this procedure:
  1. For each red number, create an R object, which will depend on data and gives the desired quantity (i.e., if data change, the value in the object changes accordingly),
  2. To force rounding a quantity to a given decimal precision, you can use `sprintf()` (as seen in Exercise 0). Do not use `round()`,
  3. To get the thousands comma separator as in "29,070", you can use `prettyNum()`,
  4. Once you have an R object for each red number formatted as desired, use `\Sexpr{}` to insert each of them inline in LaTeX (as seen in Exercise 0).

```
> n_obs <- nrow(copd) # Number of participants
> n_vars <- ncol(copd) # Number of variables
> nans <- 100*colMeans(is.na(copd)) # Percentage of missing data by variable
> min_nan <- sprintf("%.2g", min(nans)) # MIRAR # Lower percentage
> max_nan <- sprintf("%.2g", max(nans)) # Upper percentage
>
> # Vairables with less than 5% of data missing (percentage)
> less_5 <- sprintf("%.2g", 100*length(nans[nans<5])/n_vars)
>
> # Percentage of complete cases
> comp_cases <- sprintf("%.2g", 100*mean(complete.cases(copd)))
>
> # Percentage of missing data
> miss_total <-  sprintf("%.2g", 100*mean(is.na(copd)))
>
> # Total cells observed (variables*observations)
> total_cells = prettyNum(n_obs*n_vars, big.mark = ",", decimal.mark = ".")
```

"The subset used here included 342 participants and 45 variables. The range of missing data in the different variables went from 0% to 32%, with 69% of the variables having less than 5% of the data missing. Only 33% of the participants had complete data on all 45 variables. Overall, 5.4% of the information was missing; that is, 5.4% of the total of 15,390 cells ($45 \times 342 = 15,390$) had missing data."

2.2 Multiple imputation

Apply multiple imputation to copd data using 100 imputations. Explain in a reasoned manner how you decided to set the values of the arguments `method` and `predictorMatrix` when using the `mice()` function. Show your code.

```
> # Display of default imputation methods for each variable
> ini <- mice(copd, maxit=0)
> ini

## Class: mids
## Number of multiple imputations:  5
## Imputation methods:
##                id  index_barthel2v1    quo_calcpostv1         rv_prepv1        incp_fevv1
##                ""             "pmm"                ""             "pmm"             "pmm"
##       fevp_postv1          rv_tlcv1        ctnf_s_mv1    vtdvdbid_ecov1          pp_ecov1
##                ""             "pmm"             "pmm"             "pmm"             "pmm"
##          sin11_v1           vas2_v1         ic_prepv1          iam_chv1          sin15_v1
##          "logreg"             "pmm"             "pmm"          "logreg"          "logreg"
##        cil8_s_mv1           sin7_v1           sin8_v1             ch_v1         vc_prepv1
##             "pmm"          "logreg"          "logreg"             "pmm"             "pmm"
##       dtdve_ecov1          cccs1_v0      kcals_set_v0         impactsv1          monoA_v1
##             "pmm"          "logreg"             "pmm"             "pmm"             "pmm"
##  lvmass_ase_ecov1        vtit_catv1         urg_ambv0            po2_v1           ahpf_v1
##             "pmm"          "logreg"                ""             "pmm"             "pmm"
##            pro_v1           basA_v1           fc_pmv1            tad_v1            col_v1
##             "pmm"             "pmm"             "pmm"             "pmm"             "pmm"
##           sin5_v1             bc_v1       fvcp_postv1           vas1_v1           sin6_v1
##          "logreg"             "pmm"                ""             "pmm"          "logreg"
##       graudisn_v1 cpcr_mg_l_mediav1          neutA_v1           ahpl_v1        activityv1
##         "polyreg"             "pmm"             "pmm"             "pmm"             "pmm"
## PredictorMatrix:
##                  id index_barthel2v1 quo_calcpostv1 rv_prepv1 incp_fevv1 fevp_postv1
## id                0                1              1         1          1           1
## index_barthel2v1  1                0              1         1          1           1
## quo_calcpostv1    1                1              0         1          1           1
## rv_prepv1         1                1              1         0          1           1
## incp_fevv1        1                1              1         1          0           1
## fevp_postv1       1                1              1         1          1           0
##                  rv_tlcv1 ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_prepv1
## id                      1          1              1        1        1       1         1
## index_barthel2v1        1          1              1        1        1       1         1
## quo_calcpostv1          1          1              1        1        1       1         1
## rv_prepv1               1          1              1        1        1       1         1
## incp_fevv1              1          1              1        1        1       1         1
## fevp_postv1             1          1              1        1        1       1         1
##                  iam_chv1 sin15_v1 cil8_s_mv1 sin7_v1 sin8_v1 ch_v1 vc_prepv1 dtdve_ecov1
## id                      1        1          1       1       1     1         1           1
## index_barthel2v1        1        1          1       1       1     1         1           1
## quo_calcpostv1          1        1          1       1       1     1         1           1
## rv_prepv1               1        1          1       1       1     1         1           1
## incp_fevv1              1        1          1       1       1     1         1           1
## fevp_postv1             1        1          1       1       1     1         1           1
##                  cccs1_v0 kcals_set_v0 impactsv1 monoA_v1 lvmass_ase_ecov1 vtit_catv1
## id                      1            1         1        1                1          1
## index_barthel2v1        1            1         1        1                1          1
## quo_calcpostv1          1            1         1        1                1          1
## rv_prepv1               1            1         1        1                1          1
## incp_fevv1              1            1         1        1                1          1
## fevp_postv1             1            1         1        1                1          1
##                  urg_ambv0 po2_v1 ahpf_v1 pro_v1 basA_v1 fc_pmv1 tad_v1 col_v1 sin5_v1
## id                       1      1       1      1       1       1      1      1       1
## index_barthel2v1         1      1       1      1       1       1      1      1       1
## quo_calcpostv1           1      1       1      1       1       1      1      1       1
## rv_prepv1                1      1       1      1       1       1      1      1       1
## incp_fevv1               1      1       1      1       1       1      1      1       1
## fevp_postv1              1      1       1      1       1       1      1      1       1
##                  bc_v1 fvcp_postv1 vas1_v1 sin6_v1 graudisn_v1 cpcr_mg_l_mediav1 neutA_v1
## id                   1           1       1       1           1                1        1
## index_barthel2v1     1           1       1       1           1                1        1
```

```
## quo_calcpostv1       1           1       1       1           1                 1        1
## rv_prepv1            1           1       1       1           1                 1        1
## incp_fevv1           1           1       1       1           1                 1        1
## fevp_postv1          1           1       1       1           1                 1        1
##                  ahpl_v1 activityv1
## id                     1          1
## index_barthel2v1       1          1
## quo_calcpostv1         1          1
## rv_prepv1              1          1
## incp_fevv1             1          1
## fevp_postv1            1          1
```

We can see that the mice package suggest imputing initialy with de mean for continuous variables and with a logistic regression for binary ones. Further research on the dataset has shown that the variables have no deterministic relation. So we do not change the imputation methods. We have a variable (`id`) that don't have any role in the imputation process, so we change `predictorMatrix` and `methods` acording to that.

```
> # Change in parameters
> meth <- ini$method
> pred <- ini$predictorMatrix
>
> meth["id"] <- "" # do not impute "id"
> pred["id", ] <- 0 # we don't want "id" to be imputed
> pred[, "id"] <- 0 # we don't want "id" to be predictor
```

```
> # Imputation of 100 datasets
> M <- 100 # Number of imputations for each missing value
> imp <- mice(data = copd,
+             m = M,
+             method = meth,
+             predictorMatrix = pred,
+             maxit = 20,
+             printFlag = FALSE,
+             seed = 666)
> # imp£imp (output llarg)
> #summary(imp)
```

2.3 Integration of multiple imputation in cluster analysis

Reproduce the original analysis in the paper using the miclust package to integrate multiple imputation in the cluster analysis. Apply the proposed framework using the same setting than in the original analysis. To do that, follow these steps:

1. Read the help of the `getdata()` function in the `miclust` package. Then, accordingly, create a *list* of *data.frames* and pass it to the `getdata()` function. Give a detailed explanation of the data processing you did to get that *list*. Show your code.

```
> # ?miclust::getdata
> # getdata() requires the data to be a list or data.frame where the first element is the raw data:
>
>   #1- prepare raw data (all numeric and without id)
> copd_raw <- copd
> copd_raw$id <- NULL
> num_cols <- sapply(copd_raw, function(x) is.numeric(x) || is.integer(x) || is.logical(x))
> copd_raw <- copd_raw[, num_cols]
>
>   #2- prepare imputated data
> imp_list <- complete(imp, action = "all") # extracts the completed data from a mids object
> imp_list_clean <- lapply(imp_list, function(df) {
+   df$id <- NULL
+   df <- df[, num_cols]
+   return(df)
+ })
```

```
>
>   #3- create list where the first element is the raw data
> my_cluster_list <- c(list(copd_raw), imp_list_clean)
>
> # prepare the data for miclust
> imp_midata <- miclust::getdata(data = my_cluster_list)
>
> # now we check
> class(imp_midata)

## [1] "list"   "midata"

> length(imp_midata$impdata)

## [1] 100
```

The `getdata()` function needs a `data.frames` list where the first element contains the original data (with missing values) and the remaining ones correspond to multiple imputed datasets. That is why we created `my_cluster_list`, the list of data frames that we pass to `getdata()`. The imputed data sets were extracted from the `mids` object, and the same processing was applied to the original data and to each imputed data set, so that all of them have the same variables in the same order. Since k-means groups the subjects using Euclidean distances, we removed the variables that are neither numeric nor logical. We also removed the `id` variable, because it is just an identifier and does not contain any valuable clinical information. For the same reason, the `id` was not included as a predictor in the imputation.

2. Use the `miclust()` function to perform the integration of missing data in cluster analysis using the same settings as in the original analysis. Show your code and explain how and why you set all arguments in the `miclust()` function.

```
> miclust_1 <- miclust(data = imp_midata,
+                      method = "kmeans",
+                      search = "backward",
+                      ks = 2:5,
+                      usedimp = 1:100,
+                      seed = 666)

## . . . . imp   5 . . . . imp  10 . . . . imp  15 . . . . imp  20
## . . . . imp  25 . . . . imp  30 . . . . imp  35 . . . . imp  40
## . . . . imp  45 . . . . imp  50 . . . . imp  55 . . . . imp  60
## . . . . imp  65 . . . . imp  70 . . . . imp  75 . . . . imp  80
## . . . . imp  85 . . . . imp  90 . . . . imp  95 . . . . imp 100
##
##  Analysis done.

> #summary(miclust_1)
> miclust_1$kfin # optimal k (number of clusters)

## [1] 2
```

We need to perform a variable selection because adding variables can degrade the classification when the number of subjects is small compared to the numer of variables. As mentioned in the article, it is good practice that the number of subjects are at least 10 times the number of variables. In our case, the ratio is $342/44 < 10$, so a reduction of the number of variables is needed.

As in the original analysis, we use the backward search with CritCF criterion. It starts with all the variables and, at each step, removes the variable that when excluded gives the highest CritCF. This criterion allows to rank partitions with different numbers of clusters and different numbers of variables. Finally, we define `k = 2:5` where the maximum value was fixed a priori, the same way as the article. Since the backward seach is repeated for each k in each imputed data set, limiting the number of clusters reduces the computational cost and also keeps the resulting groups interpretable.

3 Results

Using `plot()` and `summary()` methods in the `miclust` package, provide plots and tables for the results similar than those shown in the original paper. Also, show a barplot for the percentage of times each variable is selected among the imputed data sets, in decreasing order. Explain how you set all arguments in all functions and methods you use. Show your code.

```
> # Distribution of the CritCF criterion over the imputed data sets, one box per
> # explored number of clusters (ks = 2:5);
> plot(miclust_1, metric = "critcf", col.critcf = "darkmagenta")
```

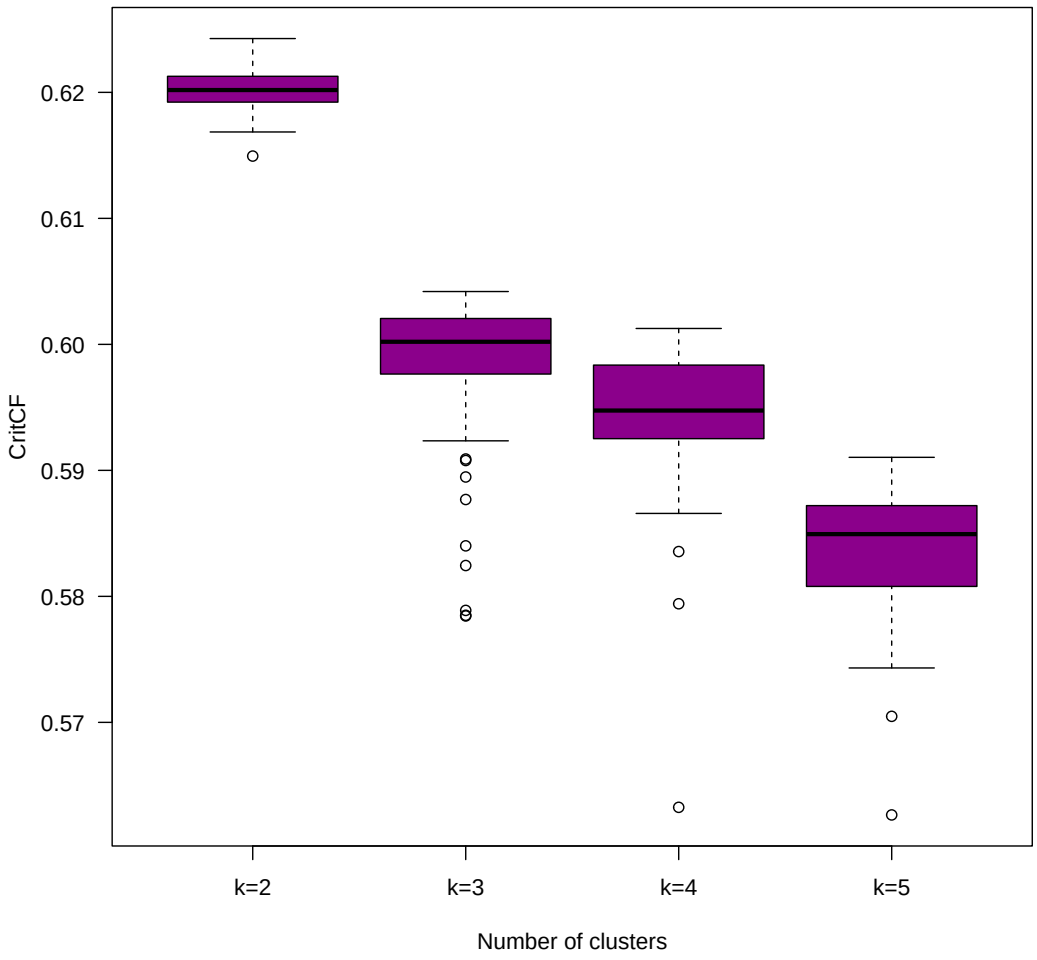


Figure 1: Boxplot of CritCF by number of clusters (k) over the 100 imputed data sets. Each box is based on 100 values; higher CritCF is preferred.

```
> # Final number of clusters selected by miclust
> kfin <- miclust_1$kfin
>
> # Percentage of appearance of each variable (metric = "varsel") for k = kfin
> plot(miclust_1, k = kfin, metric = "varsel", col.varsel = "skyblue")
```

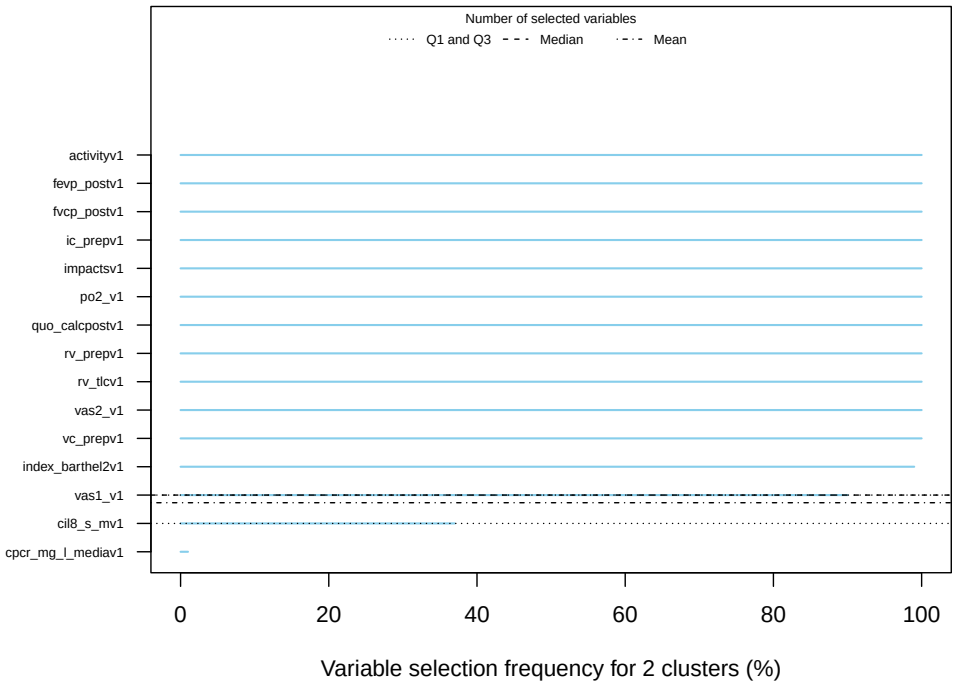


Figure 2: Variables that remained in the final set of variables in at least one of the 100 imputed data sets after the backward selection algorithm for $k = 2$, by percentage of appearance. Horizontal lines show the first and third quartiles, median and mean of the number of selected variables.

```
> # Set quantilevars = 0.5 so the numbers of final variables equals the median
> # number of variables selected across the imputed data sets.
> summary_k <- summary(miclust_1, quantilevars = 0.5)
>
> summary_kp1 <- summary(miclust_1, k = kfin + 1, quantilevars = 0.5)
```

```
> # Distribution of the probability of assignment to each cluster, for k = kfin
> a1 <- as.matrix(summary_k$allocationprobabilities)
>
> # Same distribution, for k = kfin + 1
> a2 <- as.matrix(summary_kp1$allocationprobabilities)
>
> # Numeric part, an empty row before each k block, followed by the rows of
> # the corresponding summary
> vals <- rbind(rep(NA, 5), a1, rep(NA, 5), a2)
>
> # Text part, one label per row.
> labels <- c(paste0("$k=", kfin, "$"),
+             paste0("Cluster ", seq_len(nrow(a1))),
+             paste0("$k=", kfin + 1, "$"),
+             paste0("Cluster ", seq_len(nrow(a2))))
>
> # Combine text part, first column, and numeric values into one data.frame
> tab_alloc <- data.frame(labels, vals)
>
```

```
>
> # Create the xtable object. 'digits' has 7 elements: the 1st is for the
> #automatic row names (not printed), the 2nd for the text labels (no decimals
> # apply), and the last 5 for the numeric columns (2 decimals each).
> # The caption and the label are defined here.
> alloc <- xtable(tab_alloc,
+                 digits = c(0, 0, 2, 2, 2, 2, 2),
+                 align = c("l", "l", "c", "c", "c", "c", "c"),
+                 caption = paste0("Distribution of the frequency of assignment
                         (proportion of the ", length(miclust_1$usedimp),
+                          " imputed data sets) to each cluster, for $k=",
+                          kfin, "$ and $k=", kfin + 1, "$."),
+                 label = "tab:alloc")
>
> # Column names, centered and in bold, the first one is left blank
> colnames(alloc) <- c("","\\multicolumn{1}{c}{\\bf Minimum}",
+                          "\\multicolumn{1}{c}{\\bf First quartile}",
+                          "\\multicolumn{1}{c}{\\bf Median}",
+                          "\\multicolumn{1}{c}{\\bf Third quartile}",
+                          "\\multicolumn{1}{c}{\\bf Maximum}")
>
> # Print the table un LaTeX
> print(alloc,
+       table.placement = "ht",
+       size = "footnotesize",
+       include.rownames = FALSE, # Row names are not printed, labels are in column 1
+       include.colnames = TRUE,
+       floating = TRUE,
+       sanitize.text.function = force,
+       booktabs = TRUE)
```

	Minimum	First quartile	Median	Third quartile	Maximum
$k = 2$					
Cluster 1	0.50	1.00	1.00	1.00	1.00
Cluster 2	0.57	1.00	1.00	1.00	1.00
$k = 3$					
Cluster 1	0.64	1.00	1.00	1.00	1.00
Cluster 2	0.56	1.00	1.00	1.00	1.00
Cluster 3	0.78	1.00	1.00	1.00	1.00

Table 1: Distribution of the frequency of assignment (proportion of the 100 imputed data sets) to each cluster, for $k = 2$ and $k = 3$.

```
> vars <- summary_k$selectedvariables # El nom de les variables finals seleccionades, la primera col
> summary_cluster <- summary_k$summarybycluster # És la part del summary on es detallen les propieta
>
> # % of missing values of each variable (raw data)
> miss <- summary_cluster$`%miss.` # agafem només la columna del % de valors perduts, la segona colu
>
> # Imputed data, mean of each variable by cluster (one column per cluster)
> mean_imputed <- as.matrix(summary_cluster[, grep("^mean", names(summary_cluster))]) # Matriu la qu
>
> # Majority-vote cluster of each participant
> cl <- summary_k$cluster # vector amb 342 números (en aquest cap 1 o 2, ja que són el numero de clu
>
```

```
> raw_means <- matrix(NA, nrow = length(vars), ncol = kfin) # Creem una matriu buida a on el numero
>
> F_raw <- vector(length = length(vars)) # Crem un vector buit per anar guardant els valors de F val
>
> for (i in 1:length(vars)) {
+       x <- copd_raw[[vars[i]]] # Afagem tots els valors dels participants de les variables signifi
+       fit <- lm(x ~ factor(cl)) # ajustem a un model lineal, explicar x a partir del cluster a par
+       F_raw[i] <- anova(fit)$`F value`[1] # anova() és un data.frame amb una columna F value, la q
+   for (j in 1:kfin){
+     raw_means[i,j] <- mean(x[cl == j], na.rm = TRUE) # Posem la mitjana de cada cluster en cada un
+   }
+ }
>
> # Ara fem el f value per la part imputada, la diferencia amb la part "crua" és que en lloc d'un so
>
> vector_clusters <- summary_k$clustervectors
>
> n_imp <- ncol(vector_clusters)
>
> F_imputed <- vector(length = length(vars))
>
> for (i in 1:length(vars)){
+   f_n_imp <- vector(length = n_imp)
+   for (m in 1:n_imp){
+     y <- imp_midata$impdata[[m]][[vars[i]]] # Valor de la imputació m en la variable i.
+     fit <- lm(y ~ factor(vector_clusters[, m])) # Agafem els valors de cluster que te cada individ
+     f_n_imp[m] <- anova(fit)$`F value`[1]
+   }
+   F_imputed[i] <- mean(f_n_imp)
+ }
>
> # Cluster sizes, raw data (majority vote) and median over the imputed data sets
> n_raw     <- as.vector(table(factor(cl, levels = seq_len(kfin)))) # Només hi ha un unic cluster, a
> n_imp_med <- summary_k$clusterssize[, "50%"]  # En cada imputació alguns participants poden canvia
>
>
> # Variable labels with "_" escaped for LaTeX
> labels <- gsub("_", "\\\\_", vars)
>
> tab_desc <- data.frame(labels, miss, raw_means, F_raw, mean_imputed, F_imputed)
>
> range_txt <- paste0("cluster ", seq_len(kfin), ": ",
+                     summary_k$clusterssize[, "min."], "--",
+                     summary_k$clusterssize[, "max."], collapse = "; ")
>
> desc <- xtable(tab_desc,
+                digits = c(0, 0, 1, rep(2, kfin), 2, rep(2, kfin), 2),
+                align  = c("l", "l", rep("c", 2 * kfin + 3)),
+                caption = paste0("Mean values of the ", length(vars),
+                                 " selected variables by cluster ($k=", kfin, "$). ",
+                                 "Raw data: majority-vote cluster assignment, ",
+                                 "missing values excluded. ",
+                                 "Imputed data: means over the ", n_imp,
+                                 " imputed data sets. ",
+                                 "$n$: number of participants per cluster (raw data: ",
```

```
+                                  "majority-vote assignment, imputed data: median over ",
+                                  "the imputed data sets, with minimum--maximum ",
+                                  range_txt, "). ",
+                                  "$F$: ratio of between cluster variance ",
+                                  "from a linear regression of each variable on cluster ",
+                                  "(imputed data: mean over the imputed data sets). ",
+                                  "\\% missing: percentage of participants with missing values."),
+               label = "tab:desc")
>
> # Two-level header built from computed quantities (no hand-written n)
> head_top <- paste0("& & \\multicolumn{", kfin + 1, "}{c}{\\bf Raw data} & ",
+                    "\\multicolumn{", kfin + 1, "}{c}{\\bf Imputed data} \\\\\n")
> head_sub <- paste0("\\bf Variable & \\bf \\% missing & ",
+                    paste0("\\bf Cl.~", 1:kfin, " ($n=", n_raw, "$)", collapse = " & "),
+                    " & \\bf $F$ & ",
+                    paste0("\\bf Cl.~", 1:kfin, " ($n=", n_imp_med, "$)", collapse = " & "),
+                    " & \\bf $F$ \\\\\n")
>
> print(desc,
+       table.placement = "H",
+       size = "footnotesize",
+       include.rownames = FALSE,
+       include.colnames = FALSE,            # header is written by add.to.row
+       add.to.row = list(pos = list(0),
+                         command = paste0(head_top,
+                                          "\\cmidrule(lr){3-", kfin + 3, "} ",
+                                          "\\cmidrule(lr){", kfin + 4, "-", 2 * kfin + 4, "}\n",
+                                          head_sub, "\n")),
+       floating = TRUE,
+       sanitize.text.function = force,
+       booktabs = TRUE)
```

		Raw data			Imputed data		
Variable	**% missing**	**Cl. 1** ($n = 184$)	**Cl. 2** ($n = 158$)	F	**Cl. 1** ($n = 184$)	**Cl. 2** ($n = 158$)	F
activityv1	1.2	37.21	58.86	85.26	37.01	59.02	88.56
fevp_postv1	0.0	62.90	39.09	359.64	62.87	39.08	353.71
fvcp_postv1	0.0	79.74	63.48	112.55	80.08	63.66	115.07
ic_prepv1	5.6	73.50	52.05	187.30	73.31	51.93	205.82
impactsv1	1.2	18.54	35.16	80.23	18.75	35.61	79.24
po2_v1	3.2	77.66	69.70	52.98	77.21	69.93	55.33
quo_calcpostv1	0.0	58.96	46.12	140.01	58.80	46.03	134.83
rv_prepv1	7.9	136.19	187.69	122.19	136.25	186.76	124.53
rv_tlcv1	7.9	50.26	63.59	266.57	50.43	63.59	280.59
vas2_v1	1.2	4.23	6.34	56.62	4.24	6.43	58.50
vc_prepv1	5.6	78.00	60.59	160.56	77.87	60.11	180.36
index_barthel2v1	2.6	99.89	98.32	14.83	99.93	98.31	15.99
vas1_v1	1.2	1.99	3.99	58.60	1.96	4.05	59.22

Table 2: Mean values of the 13 selected variables by cluster ($k = 2$). Raw data: majority-vote cluster assignment, missing values excluded. Imputed data: means over the 100 imputed data sets. n: number of participants per cluster (raw data: majority-vote assignment, imputed data: median over the imputed data sets, with minimum–maximum cluster 1: 179–189; cluster 2: 153–163). F: ratio of between cluster variance from a linear regression of each variable on cluster (imputed data: mean over the imputed data sets). % missing: percentage of participants with missing values.

```
> # Frequency of selection of each variable across the imputed data sets
> y   <- getvariablesfrequency(miclust_1)
>
> # Sort the variables by decreasing percentage of selection
> y_ord <- order(y$percfreq, decreasing = TRUE)
> perc <- y$percfreq[y_ord]
> name <- y$varnames[y_ord]
>
> # Bars drawn as thick vertical lines (type = "h", lwd = 20).
> plot(perc, type = "h", main = "", xlab = "", ylab = "Percentage of times selected",
+      xlim = 0.5 + c(0, length(perc)), ylim = c(0, 100),
+      lwd = 20, col = "skyblue", xaxt = "n")
>
> # Variable names below each bar, las = 2 draws them vertically, and
> # cex.axis = 0.9 shrinks them slightly so that the long names fit
> axis(1, at = 1:length(name), labels = name, las = 2, cex.axis = 0.9)
```

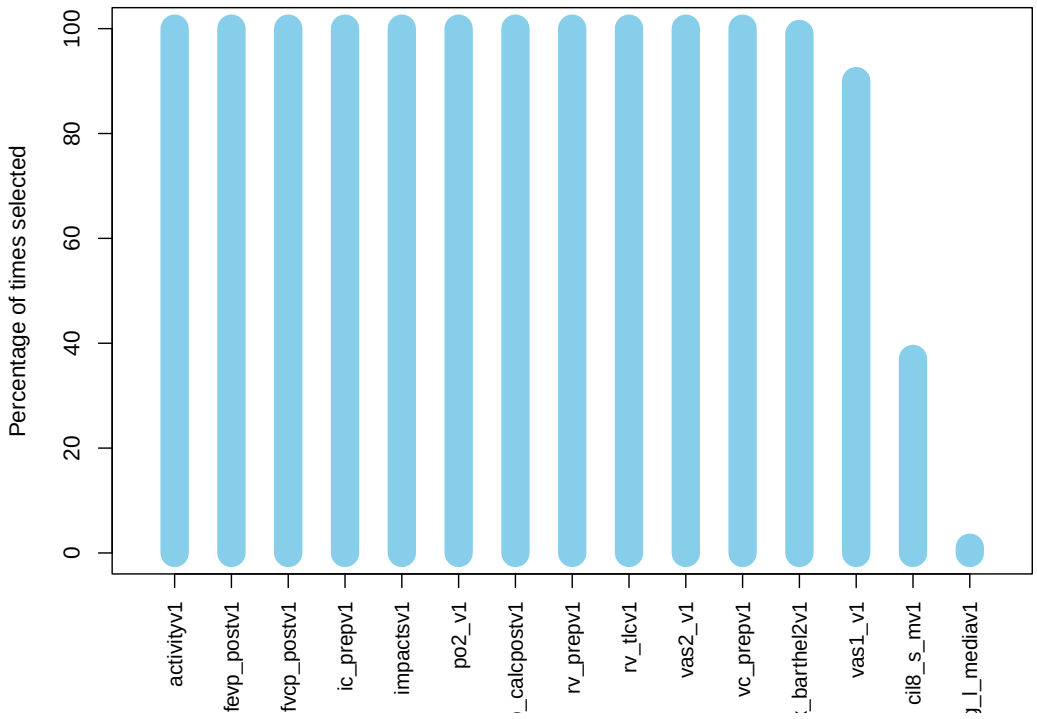


Figure 3: Percentage of the 100 imputed data sets in which each variable was selected by the backward algorithm for $k = 2$, in decreasing order.

```
> # This code is adapted from the miclust README, which can be accessed at
> # https://cran.r-project.org/web/packages/miclust/readme/README.html
```

4 Report

Write a brief report (400 words maximum) describing the following issues, and citing tables and figures:

- The final number of clusters and their sizes and uncertainty (cite your figure(s) and table(s)).
- The agreement when relabeling the clusters.
- The final subset of clustering variables and uncertainty.

- The probability of assignment to each cluster and uncertainty.

To find the number of clusters, we applied the backward search with the CritCF criterion to each of the 100 imputed data sets. Figure 1 shows that even the worst value for $k = 2$ is above the best value for $k = 3$, so $k = 2$ was the optimal number of clusters in all the imputed data sets. Therefore, there is practically no uncertainty in the choice of k. After the assignation, the clusters contain 184 and 158 subjects.

To prevent a potential problem caused by the automatic labeling of the clusters, the clusters were relabeled taking the first data set as a reference and evaluating them with Cohen's kappa, same as the article. The agreement after relabeling is good, with a median Cohen's kappa of 0.97.

For $k = 2$, 15 of 33 variables candidates were selected in at least one data set(Figure 2 and 3). The final number of selected variables is 13, with 11 of those that are in all the imputed data sets. There is a clear drop in frequency right after the last selected variable (from 90% to 37%). Also, we see that the selected variables would be the same if we kept those appearing in more than 50% of the data sets. Therefore, the variable selection is very robust to the missing data.

Each subject was assigned to the cluster to which they were allocated most often across the imputed data sets. In table 1 we can see how this frequency is distributed in $k = 2$ and $k = 3$. For $k = 2$, the median probability is 1.00 and 1.00 in both clusters, so most subjects are always assigned to the same cluster. For $k = 3$, the assignment is also very stable, and this is probably because the selected variables have few missing data, so the imputed data sets are very similar. Given this, we see that a stable assignment does not necessarily mean that the clusters are adequate, we have already seen that the choice of $k = 2$ is supported by the CritCF criterion. Table 2 shows that cluster 2 groups the subjects with more severe disease with worse lung function, more air trapping, lower arterial oxygen and worse scores in the questionnaires. We also see that results with raw and imputed data are very similar.